



DCM Boost Converter Project

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1 Introduction

This project details the design of an open-loop power converter interfacing a 3-cell series (3S) Li-ion battery bank with a Dell laptop load. Following the assigned parameters for Team 5, a Boost topology operating strictly in Discontinuous Conduction Mode (DCM) was implemented. The primary objective is to ensure stable voltage regulation across varying input states of charge and load demands, while maintaining minimum costs.

2 System Specifications

The converter is designed to satisfy specific nominal and extreme boundary conditions for the battery source and the load.

2.1 Input Battery Source

The energy source is a nominal 11.1V Li-ion battery pack, configured using 3 cells in series (3S). To simulate real-world battery discharging behavior, the design accounts for a $\pm 20\%$ variation in the input voltage:

- **Nominal Input:** $V_{nom} = 11.1V$
- **Minimum Input (Depleted):** $V_{min} = 8.88V$
- **Maximum Input (Fully Charged):** $V_{max} = 13.32V$

2.2 Output Load Conditions

The nominal load is defined by a standard Dell laptop charger outputting 20V at 4.5A. Modeled as a resistor in simulation, the converter must maintain this 20V output while the load varies between 50% and 120% of the nominal current:

- **Nominal Load (100%):** $I_{out} = 4.5A \implies R_{nom} = 4.44 \Omega$
- **Maximum Load (120%):** $I_{out} = 5.4A \implies R_{min} = 3.70 \Omega$
- **Minimum Load (50%):** $I_{out} = 2.25A \implies R_{max} = 8.88 \Omega$



Figure 1: Standard Dell 90W (20V, 4.5A) laptop charger used as the load baseline.

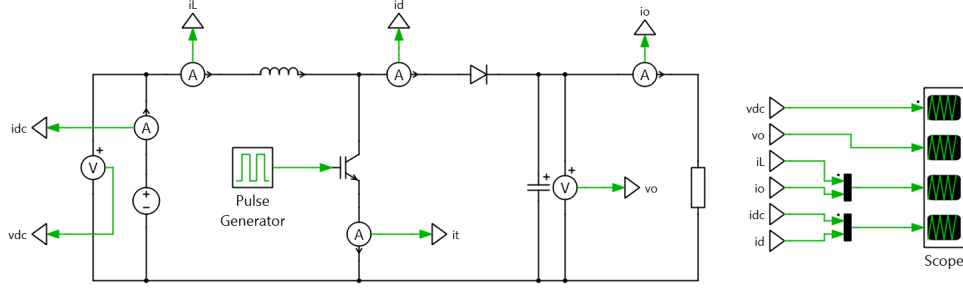


Figure 2: Complete PLECS model showing the input voltage, components, pulse generator, and resistive load.

3 Design and Component Selection

To optimize the size and cost of the passive components, the switching frequency was set to a standard $f_s = 100$ kHz.

3.1 Inductor Selection

To maintain DCM across all scenarios, the inductor must be smaller than the critical inductance (L_{crit}). The worst-case scenario for entering Continuous Conduction Mode (CCM) occurs at the minimum input voltage and maximum load. First, the CCM boundary duty cycle (D) is determined:

$$(1) \quad D = 1 - \frac{V_{in,min}}{V_{out}} = 1 - \frac{8.88}{20} = 0.556$$

The critical inductance at this boundary is calculated as:

$$(2) \quad L_{crit} = \frac{R_{min}}{2 \cdot f_s} \cdot D \cdot (1 - D)^2 \approx 2.03 \mu\text{H}$$

Selection: An inductor of $L = 1.5 \mu\text{H}$ was selected to provide a safe margin and ensure deep DCM operation.

3.2 Capacitor Selection

The output voltage ripple (ΔV_{out}) is restricted to a maximum of 4%, which equates to 0.8V. The worst-case ripple occurs under maximum load and maximum duty cycle conditions:

$$(3) \quad C_{min} = \frac{I_{out,max} \cdot D}{f_s \cdot \Delta V_{out}} \approx 37.53 \mu\text{F}$$

Selection: A standard capacitor value of $C = 100 \mu\text{F}$ was selected to provide a significant safety margin against non-ideal component behaviors.

3.3 Semiconductor Selection (MOSFET and Diode)

To safely operate in open-loop DCM, the active components must withstand the absolute peak inductor current (I_{peak}) and the maximum output voltage. The worst-case peak current occurs at the minimum input voltage and maximum load ($V_{in} = 8.88\text{V}$, $D = 0.48$), resulting in a calculated $I_{peak} \approx 28.3\text{A}$.

- **MOSFET:** The RS1G300GNTB was selected because its drain-to-source breakdown voltage ($V_{DS} \geq 30\text{V}$) safely exceeds the 20V output, and its pulsed drain current rating easily accommodates the 28.3A switching spikes.
- **Power Diode:** The SBX3040-3G Schottky diode was chosen for its low forward voltage drop (to maximize efficiency) and its reverse voltage rating ($V_R \geq 30\text{V}$), which safely prevents breakdown when the MOSFET is active.

4 Bill of Materials (BOM)

The components were selected from real-world distributors to withstand the worst-case peak currents and voltages while ensuring a highly cost-effective design.

Table 1: Bill of Materials for the DCM Boost Converter

Component	Part Number	Key Parameters	Price (USD)
MOSFET	RS1G300GNTB	$V_{DS} \geq 30\text{V}$, $I_D \geq 30\text{A}$	\$0.86
Power Diode	SBX3040-3G	$V_R \geq 30\text{V}$, $I_F \geq 30\text{A}$	\$0.50
Inductor	AWIM1608065D-R47MTF	$1.5\ \mu\text{H}$	\$0.10
Capacitor	860040573004	$100\ \mu\text{F}$	\$0.12
Total Cost:			\$1.58



5 Simulation Results and Analysis

To verify the theoretical design and ensure the boost converter operates exclusively in Discontinuous Conduction Mode (DCM) under all conditions, open-loop simulations were conducted using PLECS. The system was tested across three extreme corner cases.

5.1 Case A: Worst-Case Heavy Load and Minimum Input Voltage

Analysis: This scenario represents the boundary condition where the converter is most likely to slip into Continuous Conduction Mode (CCM). As observed in the inductor current waveform, the current successfully drops to 0A and remains zero for a portion of the switching period, confirming that the selected inductor ($1.5 \mu\text{H}$) guarantees DCM operation even under the heaviest load. The output voltage is successfully regulated at 20V with a ripple well within the 4% limit ($< 0.8\text{V}$).

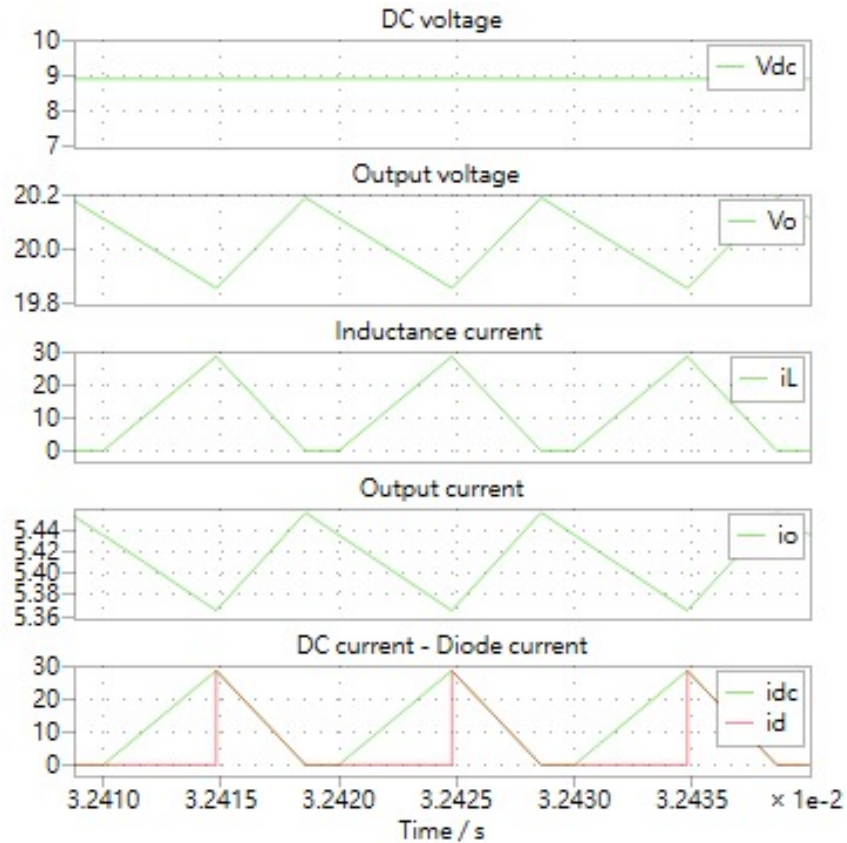


Figure 3: Simulation results for Case A ($V_{in} = 8.88\text{V}$, $R = 3.70 \Omega$, Duty Cycle = 0.48). Top to Bottom: Input DC Voltage, Output Voltage, Inductor Current, Output Current, and Diode Current.

5.2 Case B: Nominal Operating Condition

Analysis: This case demonstrates the standard operation of the converter charging the laptop at its rated power (90W). With the theoretical duty cycle applied, the output voltage centers precisely at 20V. The inductor current exhibits standard DCM characteristics, with predictable peak currents that inform the proper selection of the physical MOSFET and diode.

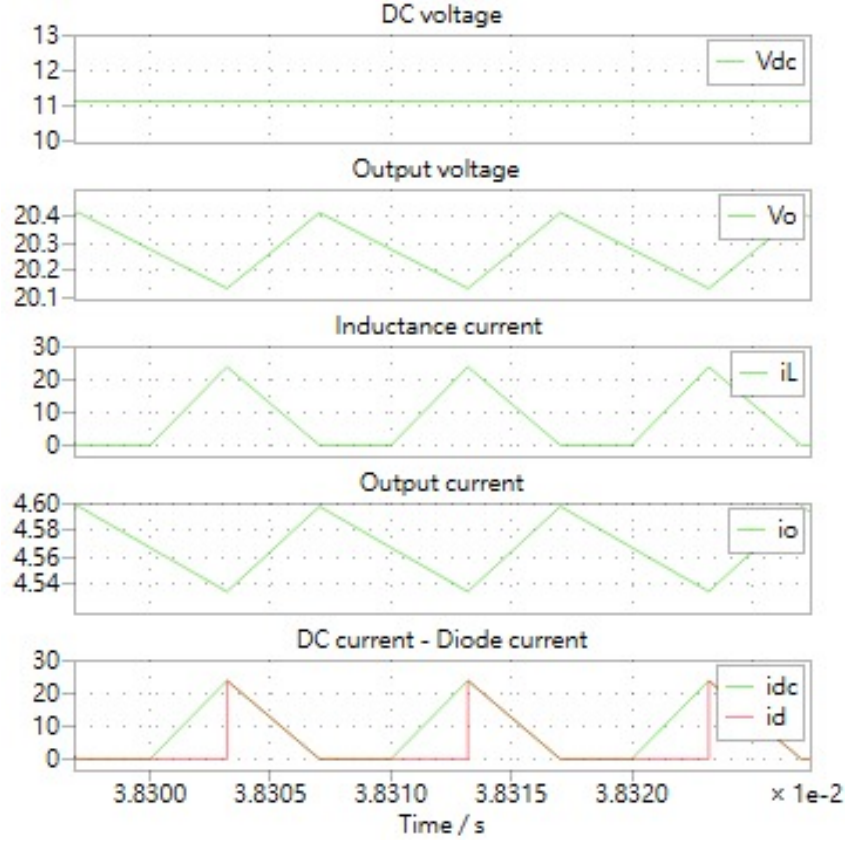


Figure 4: Simulation results for Case B ($V_{in} = 11.1\text{V}$, $R = 4.44\ \Omega$, Duty Cycle = 0.32). Top to Bottom: Input DC Voltage, Output Voltage, Inductor Current, Output Current, and Diode Current.

5.3 Case C: Worst-Case Light Load and Maximum Input Voltage

Analysis: In DCM, open-loop converters are highly susceptible to overvoltage during light load conditions. This test proves that by drastically reducing the duty cycle to 0.16, the system prevents excessive energy transfer to the output capacitor. The inductor current peaks are significantly lower, and the output voltage remains safely stabilized at 20V without any voltage runaway.

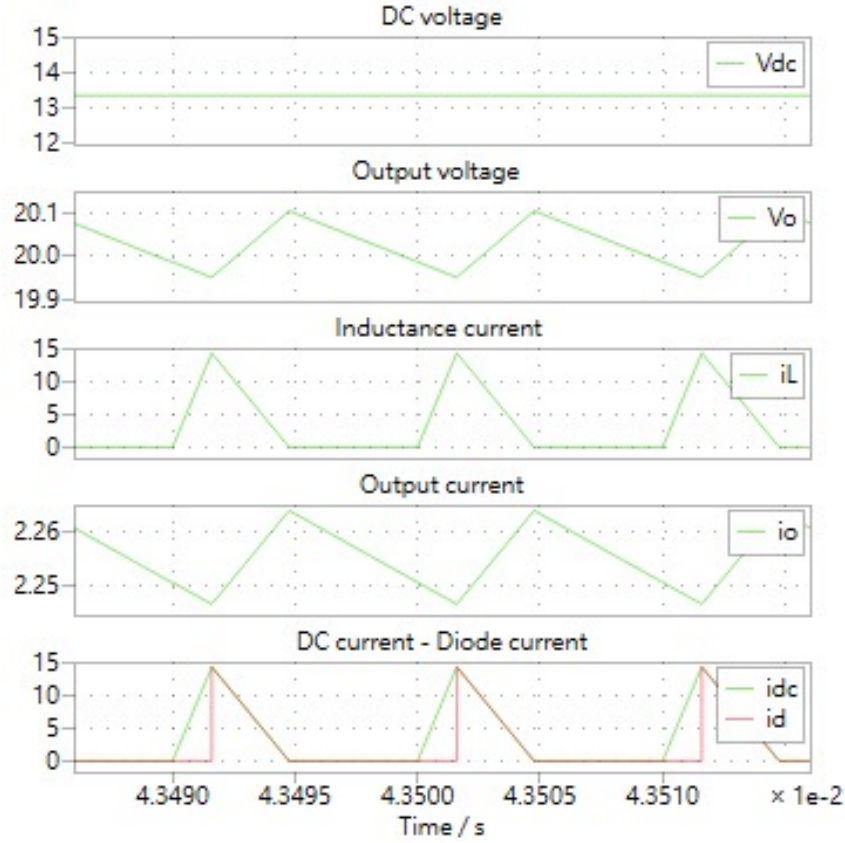


Figure 5: Simulation results for Case C ($V_{in} = 13.32\text{V}$, $R = 8.88\ \Omega$, Duty Cycle = 0.16). Top to Bottom: Input DC Voltage, Output Voltage, Inductor Current, Output Current, and Diode Current.

6 Conclusion

The proposed design successfully meets all project requirements for a 20V laptop power supply. By utilizing a $1.5\ \mu\text{H}$ inductor and a $100\ \mu\text{F}$ capacitor, the converter remains safely in Discontinuous Conduction Mode while keeping the voltage ripple well below the 4% threshold across all load and input variations. Furthermore, the final design is highly cost-efficient, with a total component cost of under \$2.00.